Computing Fundamentals
EC-110

Chapter 2A

Using the Keyboard And Mouse

Overview Of Input Devices

- Used to *input data to computer*
- If *computer is brain*, *input devices are sensory organs*
  - *Keyboard and Mouse*
  - *Devices for Hand*

    > Pens > Touch screens

    > Game controllers
  - *Optical Input Devices*

    > Bar Code Readers > Image Scanners and OCRs (Optical Character Recognition)
  - *Audio Visual Input Devices*
    - Microphones
    - Other Audio Inputs (Musical Instruments)
    - Video Inputs
    - Digital Cameras

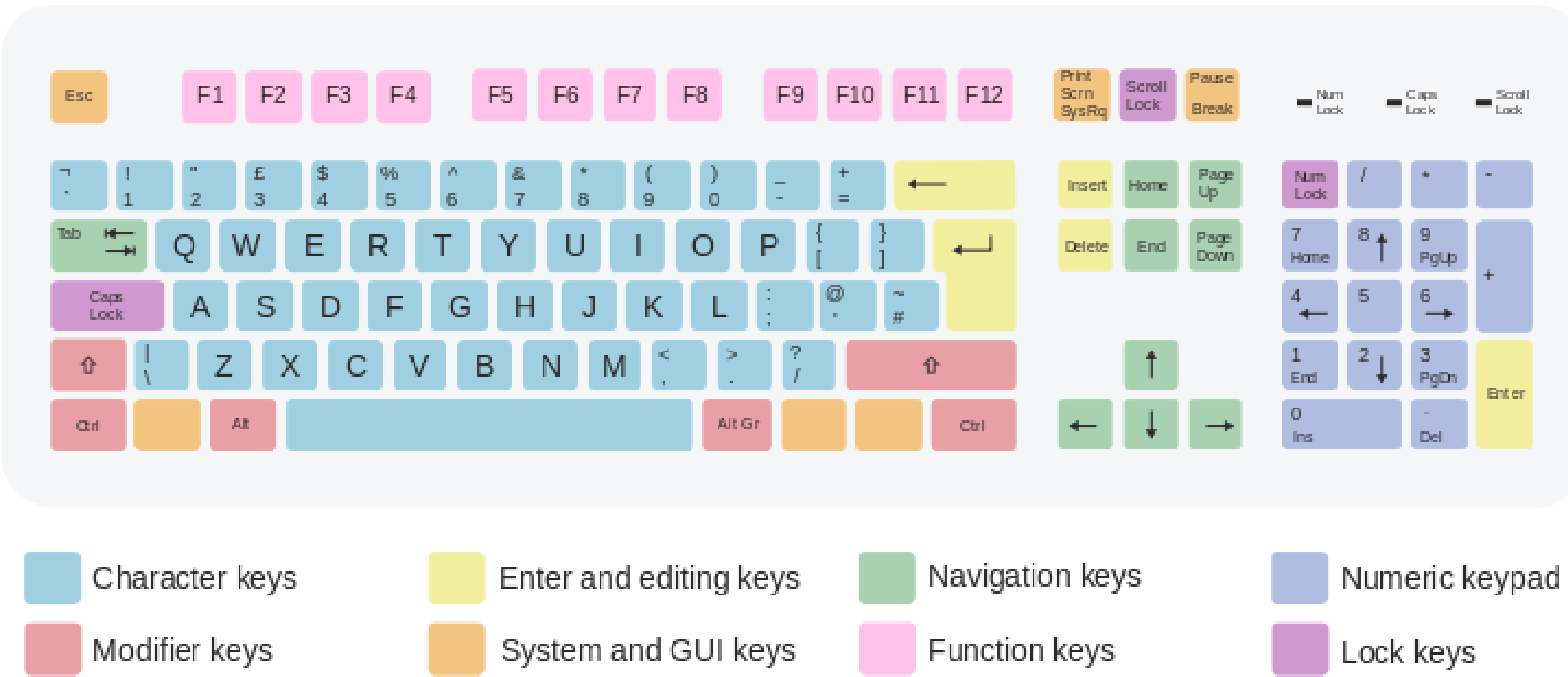
The Keyboard

- The most common input device
  - Must be proficient with keyboard
  - Skill is called keyboarding
  - Has 100+ keys.
  - Each key sends different signal to CPU

Standard Keyboard Layout

- Character or Alphanumeric Keys
- Modifier Keys
- Numeric Keypad
- Function Keys
- Navigation or Cursor Movement Keys
- Special Purpose Keys

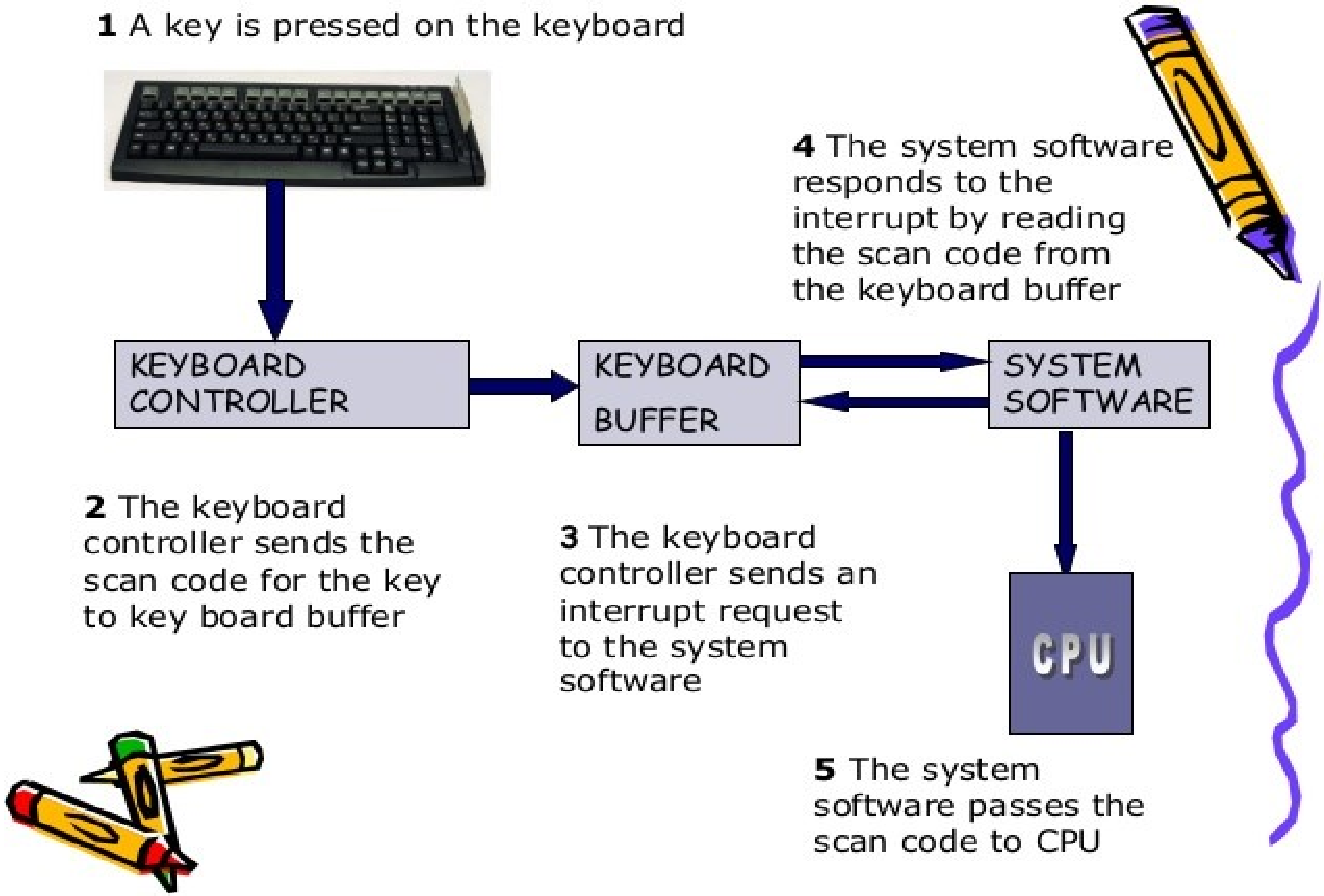
Standard Keyboard Layout



The Keyboard

- How keyboard works
  - *Keyboard controller detects a key press*
  - Places *a code* into part of its memory called keyboard buffer
  - *Controller sends code to the CPU* through system software
    - Code represents the key pressed
  - Controller notifies the operating system
  - *Operating system responds*
  - Controller repeats the letter if held too long
  - Setting for it is called ‘repeat rate’

How PC accept Input from Keyboard ?

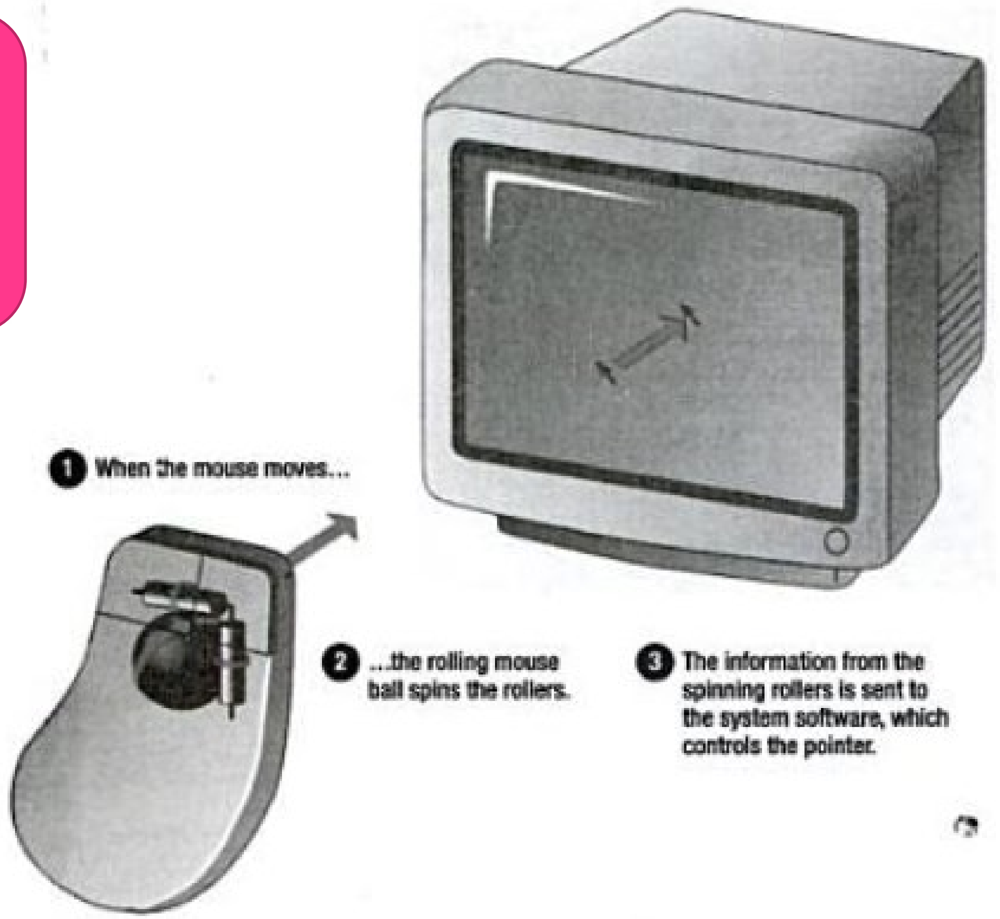


The Mouse

- All modern computers have a variant
- *Allows users to select objects*
  - Pointer moved by the mouse
  - Pointer is an on screen object
- *Has a rubber ball connected to rollers and sensors*
- *Sensors* send signals to computer
- *Rubber ball* determines distance, direction and speed
- The ball often requires cleaning

How PC accepts Input from Mouse ?

Prepared till this slide (14.10.2020)



The Mouse

- Optical mouse
  - Light shown onto mouse pad
  - Reflection determines speed and direction
  - Requires little maintenance

The Mouse

- Interacting with a mouse
  - Actions involve pointing to an object
  - Clicking selects the object
  - Clicking and holding drags the object
  - Releasing an object is a drop
  - Right clicking activates the shortcut menu
  - Modern mice include a scroll wheel

The Mouse

- Benefits
  - Pointer positioning is fast
  - Menu interaction is easy
  - Users can draw electronically

Variants of the Mouse

- Trackballs
  - Upside down mouse
  - Hand rests on the ball
  - User moves the ball
  - Uses little desk space

Variants of the Mouse

- Track pads
  - Stationary pointing device
  - Small plastic rectangle
  - Finger moves across the pad
  - Pointer moves with the pointer
  - Popular on laptops

Variants of the Mouse

- Track point
  - Little joystick on the keyboard
  - Move pointer by moving the joystick

Ergonomics and Input Devices

- Ergonomics
  - Study of human and tool interaction
  - Concerned with physical interaction
  - Attempts to improve safety and comfort

Ergonomics and Input Devices

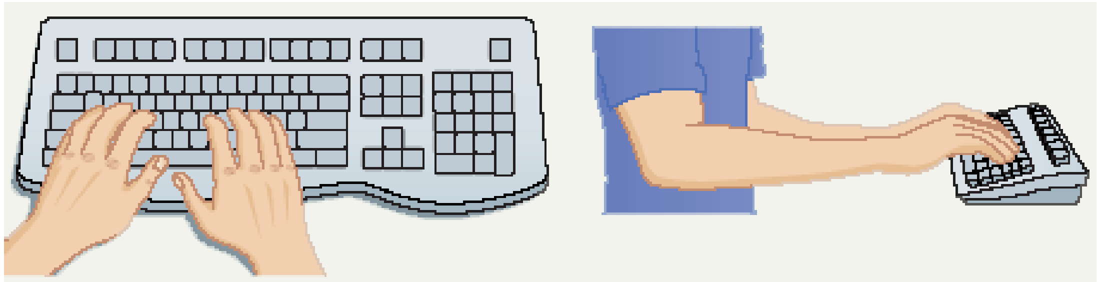
- Repetitive Strain Injury (RSI)
  - Caused by continuous misuse of the body
  - Many professions suffer from RSI
- Carpal Tunnel Syndrome
  - Carpal tunnel is a passage in the wrist
  - Holds nerves and tendons
  - Prolonged keyboarding swells tendons

Ergonomics and Input Devices

- Office hardware suggestions
  - Office chairs should have
    - Adjustable armrests and height
    - Armrests
    - Lower back support
  - Desks should have
    - Have a keyboard tray
    - Keep hands at keyboard height
    - Place the monitor at eye level

Ergonomics and Input Devices

- Techniques to avoid RSI
  - Sit up straight
  - Have a padded wrist support
  - Keep your arms and wrist straight
  - Keyboard properly
  - Take frequent breaks



Class Participation

- COMPUTER stands for??
- Different PORTS in your System??

Computer Ports

- A **computer port** is a connection point or interface between a computer and an external or internal device.

- Internal ports may connect such devices as hard drives and CD ROM or DVD drives

- External ports may connect modems, printers, mice and other devices.

  - Serial (Mice , Modems).
  - Parallel (Printers).
  - SCSI (hard-drive, other drives, up-to 7 devices).
  - USB ports.

Chapter 2B

Inputting Data In Other Ways

Devices for the Hand

- Pen based input
  - Tablet PCs, PDA.
  - Pen used to write data
  - Pen used as a pointer
  - As a tapping device
  - Handwriting recognition
  - On screen keyboard
  - Used for short notes taking, inputting signatures, delivering orders

Devices for the hand

- Touch Screens
  - Sensors determine where finger points
  - Sensors create an X,Y coordinate
  - Usually presents a menu to users
  - Found in cramped or dirty environments

Devices for the hand

- Game Controllers
  - Enhances gaming experience
  - Provide custom input to the game
  - Modern controllers offer feedback
  - Joystick
  - Game pad

Optical Input Devices

- Allows the computer to see input

- Bar Code Readers

  - Converts bar codes to numbers

  - Computer find number in a database

  - Works by reflecting light
    - Amount of reflected light indicates number

Optical Input Devices

- Image Scanners
  - Converts printed media into electronic
  - Reflects light off of the image
  - Sensors read the intensity
  - Filters determine color depths

Optical input devices

- Optical Character Recognition (OCR)
  - Converts scanned text into editable text
  - Each letter is scanned
  - Letters are compared to known letters
  - Best match is entered into document
  - Rarely 100% accurate

Audiovisual Input Devices

- Microphones
  - A sound card is a device which translates analogue audio signals to digital (a process called digitizing) and vice versa
  - Used to record speech
  - Speech recognition
    - “Understands” human speech
    - Allows dictation or control of computer
    - Matches spoken sound to known phonemes
    - Enters best match into document

Audiovisual Input Devices

- Musical Instrument Digital Interface (MIDI)
  - Connects musical instruments to computer
  - Digital recording or playback of music
  - Musicians can produce professional results

Audiovisual Input Devices

- Video Input
  - Uses PC video Camera
  - Digitizes the image by breaking it into pixels
  - Its color and other characteristics are stored as digital code
  - Usually now Webcam is used
  - Using Video capture cards other video devices can also be connected

Audiovisual Input Devices

- Digital cameras
  - Captures images electronically
  - No film is needed
  - Image is stored as a JPG file
  - Memory cards store the images
  - Used in a variety of professions